

DATASET STRUCTURE:

The dataset is composed of numerous drone shot images mainly capturing views of urbanized areas. Images are taken at **different angles** and **perspectives**, capturing various types of objects with **varying concentration** of each. Some images are mainly holding lorries, vehicles while others are vastly composed of crowd gatherings, where pedestrians and people are populating major parts of the images.

The dataset has mainly four folders - test challenge, test dev, train and validation. Each has an image and label folder. The images are mainly of **JPG** format. For validation, there are 548 images preserved. The training set has 6471 images while the test has 1610 images.

The **Labels** folder holds the class, coordinates, width and height of the bounding box for all the objects in an image. These labels are all **normalized from 0 to 1**, so every label attribute is circumscribed to depict a value in this range. This is basically following the traditional YOLO format for the labels.

There are ten classes for classification namely, pedestrian, people, bicycle, car, van, truck, tricycle, awning-tricycle, bus and motor.

There are images that have a good balance of vehicles and humans, each having a similar count with uniform scattering. Almost all images have very close up captures of objects, exposing as much detail and structures as the shape of a car, wheel designs, headlights etc. Meanwhile, concerned objects also appear as **distant** in images, where only a few pixels hardly make up the true shape of the objects, offering a greater challenge to models being trained on this dataset. Some images are also **blurry**, which disrupt the general shape and greatly hide aforementioned details.

Background inconsistency is also prominent - some images are taken from a higher angle with the drone stooping down, hence mainly capturing the

concrete floor, while other angles expose greenery plants and trees, creating a more greenish background. Some settings are completely different where shots are taken in interiors such as basketball courts, creating reddish backgrounds.

PREPROCESSING AND AUGMENTATION STEPS:

One of the first steps to clean the dataset is to ensure every label has exactly five attributes - class, x-coordinate, y-coordinate, width and height. Any label that violates this condition is discarded along with their images. Additionally, it is imperative to determine the size of the bounding boxes once the images get scaled down to 640 x 640 pixels. For this assessment, we stuck to training the images in this resolution. Scaling down any image will also diminish the sizes of the bounding box areas. As a result, we ensure that the width and height is **greater than some threshold**, ensuring that the objects within the bounding boxes are somewhat big in size for the model to learn the features of an object.

Next, in order to make the images more consistent in terms of lighting, we use **CLAHE** to perform local histogram equalization. This creates tiles in the image and histogram equalization is performed in every tile independently to **evenly distribute** pixels across the entire pixel value range, solidifying contrast.

Afterwards, augmentation on these images are also performed. This includes adding Gaussian noise, randomly rotating images, tweaking brightness, hue saturation values. This introduces a larger diversity and makes the model learn the objects instead of **memorizing** their general patterns.

At last, we tile an image with an overlapping threshold. This operation is similar to **SAHI Tiling** to preserve the size of objects when scaled down to 640 x 640 pixels. Since bounding boxes can fall in between two adjacent tiles, we perform a check with a **visibility threshold** to determine which tile

the bounding box belongs to. If the bounding box area that is left after getting clipped by the tile is less than a certain fraction of the total, the bounding box is not considered to belong in that tile.

CHALLENGES IN THE DATASET:

The dataset has a significant class imbalance as depicted below:

- pedestrian: 79337
- people: 27059
- bicycle: 10480
- car: 144867
- van: 24956
- truck: 12875
- tricycle: 4812
- awning-tricycle: 3246
- bus: 5926
- motor: 29647

To resolve this, image oversampling could have been done to accentuate the importance of rarer classes by simply repeating images that contain those rare classes. Since the dataset is already very big, this duplication immensely increased the size of the dataset beyond Kaggle's space accommodation.

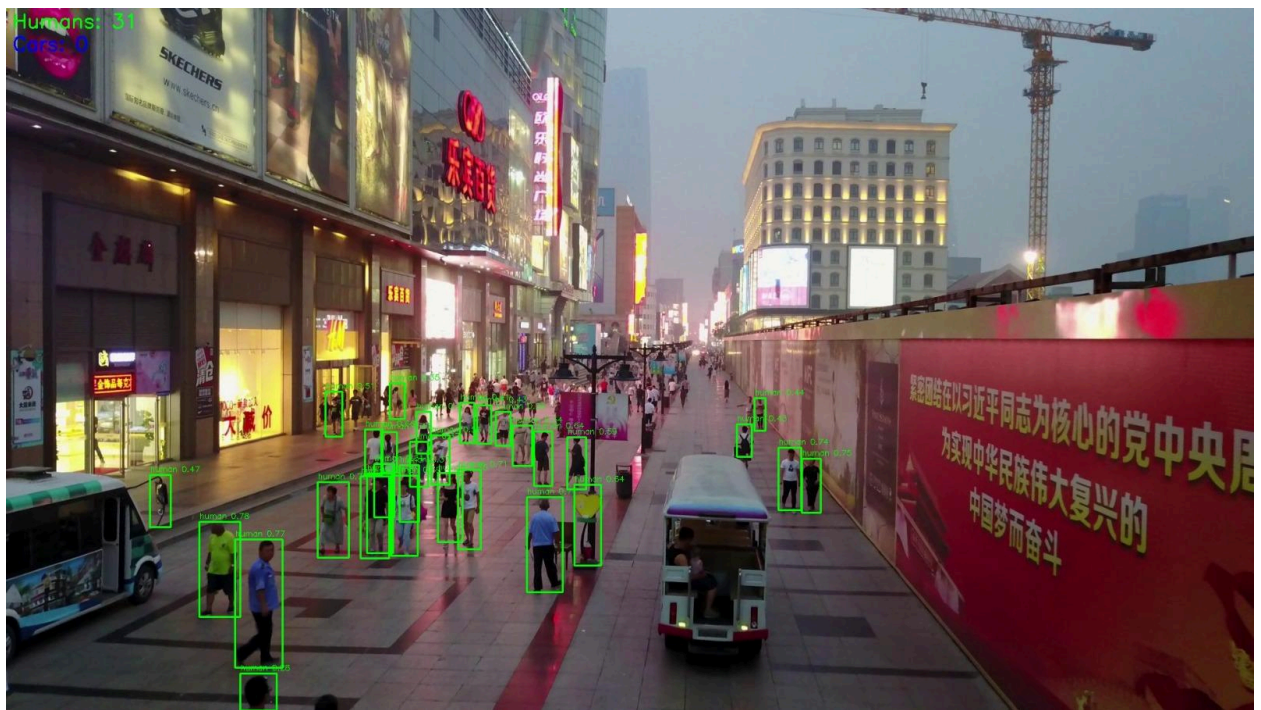
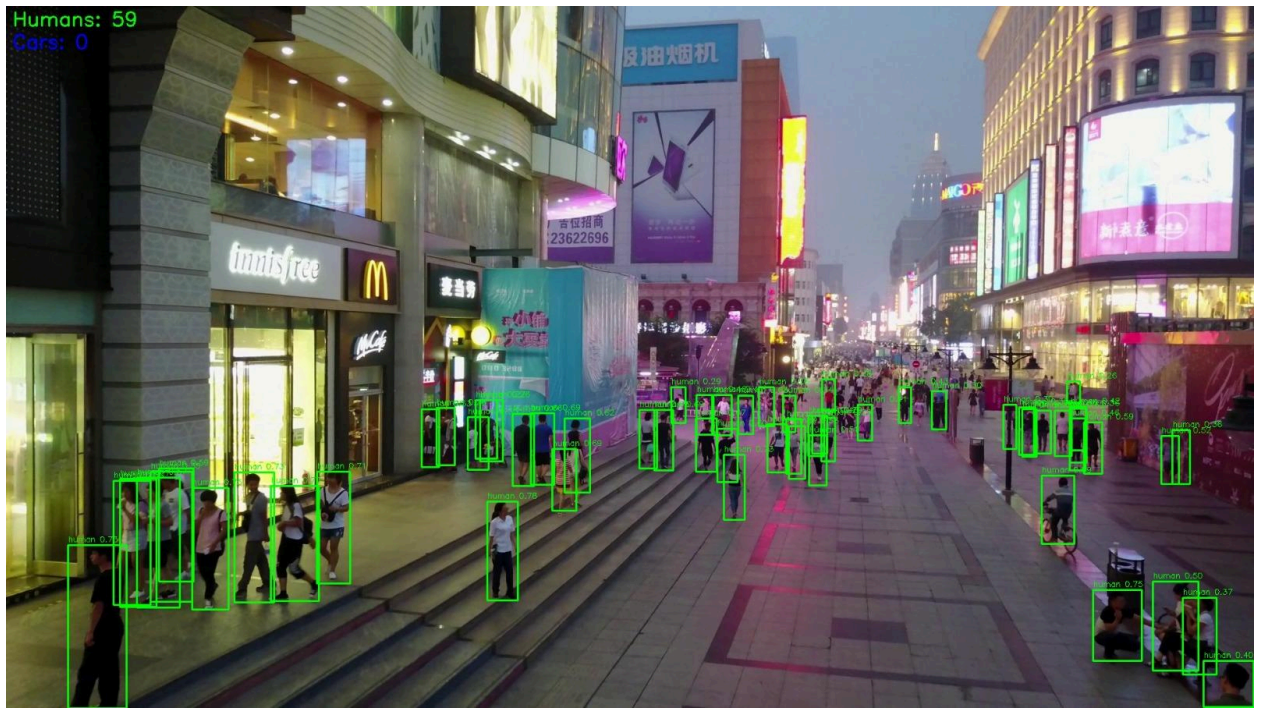
The images are also taken at different times of the day, breaking the consistency of **lighting conditions** and creating very long and short shadows based on the lighting position. The images are also taken at different times of the day, breaking the consistency of **lighting conditions** and creating very long and short shadows based on the lighting position.

RESULTS:

After 50 epoch, inference has been run on the test dataset and the following metrics have been obtained:

```
Precision   : 0.4338
Recall      : 0.3108
mAP50      : 0.2804
mAP50-95   : 0.1584
DETECTION COUNTS
=====
Images processed : 1610
Total humans     : 7755
Total cars      : 23802
Total time      : 105.73s
```

Below are some images of predicted outputs, showcasing counting and labelling on bounding boxes:





DISCUSSION:

The model is performing relatively well considering only being finetuned through 50 epochs. Despite lower lighting conditions, the model has been still able to detect a majority (if not all) objects in the third image.

However, the model is still unfamiliar with some vehicles. In the second image, there are two transports that have a different structure than regular cars which are left unlabelled by the model. This either means that the training set of the dataset does not take into account all types of vehicles.

Main challenges were regarding augmenting the dataset which continuously increased space usage in Kaggle's given space. This prevented implementation to handle class imbalance which is a probably a significant reason for the model to not necessarily show satisfactory metrics.